Capstone Progress Update 3: Weeks 5 and 6

An Automated Data Pipeline for the German Electricity Market

1. Where I stand

The project is complete. In these two weeks I answered the last two questions: a forecast of tomorrow's prices and a check of how reliably my pipeline runs. The pipeline now runs on a daily schedule, loads fresh data every day and passes all of its quality checks. What is left is to finish the written report and record the video.

2. What I did

Week	What I did	How it went
5	Built a price forecast with gradient boosting and a simple baseline to compare against	The model beats the baseline, see below
5	Tested it on the newest 8 weeks, which it had never seen	A fair test, with no peeking at future data
6	Answered question 5 from the run log: freshness, completeness and failed runs	Healthy now, see below
6	Finished the dashboard and the README	Done
6	Started the draft of the final report	In progress

3. Question 4: can I forecast tomorrow's hourly prices better than a simple rule?

I forecast every hour of day D at noon on D-1, which is the auction deadline. The model only uses what is known at that moment: prices up to the end of D-1, power and weather values from two days earlier, and the calendar. I trained it on January 2023 to August 2026 (32,111 hours) and tested it on the newest 8 weeks, 31 August to 25 October 2026 (1,344 hours).

Method	MAE (EUR/MWh)	RMSE (EUR/MWh)
Same hour last week (baseline)	63.4	84.1
Gradient boosting	47.4	60.5

The model's average error is 47.4 EUR/MWh against 63.4 for the simple rule, a 25% reduction. It was better in 7 of the 8 test weeks. Its largest errors are at 19:00 (83.9 EUR/MWh), the hour with the evening price peak.

Why it is not better still. Real traders use wind, solar and load forecasts for the day they are trading. My model only sees actual values from two days earlier, because forecasts are not in my pipeline. Adding the day-ahead generation forecasts from SMARD would be the most promising next step. Gas and CO_2 prices are also missing.

4. Question 5: does my pipeline run reliably?

This is a snapshot from 25 October 2026, 09:13 UTC. The run statistics and the recent completeness cover the last 30 days.

Freshness

Source	Last successful load (UTC)	Newest data (UTC)	Lag (hours)
Prices	25 Oct 09:12	25 Oct 22:00	-12.8
Power	25 Oct 09:12	25 Oct 03:00	+6.2
Weather	25 Oct 09:13	24 Oct 23:00	+10.2

A negative lag means the data reaches into the future, which is normal for day-ahead prices.

- **Completeness.** In the last 30 days, 100.00% of hours have a value for prices, power and temperature. Power and weather are judged up to now minus their usual publication delay (12 and 48 hours). The lowest monthly value over the whole history is also 100.00%.
- **Runs.** 7 runs have finished in total, all of them in the last 30 days: 6 succeeded and 1 was abandoned. Later runs recovered.
- **Quality checks.** The latest run passes all 19 checks.

Verdict. The pipeline is healthy now: the data is fresh and complete, and all quality checks pass. One stage was abandoned in the last 30 days and is listed in the failures file. This answer becomes more meaningful the longer the daily schedule runs, because each run adds a row to the run log.

5. Looking back at all five questions

Question	Main result
1. Prices by hour, weekday and season	Cheapest at 13:00 (50.8), most expensive at 19:00 (160.9 EUR/MWh); weekends 27% cheaper
2. Effect of renewable share	+10 points of share goes with about 16 EUR/MWh lower price, stable over time
3. Best time for flexible use	Cheap and green hours coincide at midday; moving a 3-hour load saves 70% per run
4. Forecast of tomorrow's prices	Gradient boosting is 25% better than the same-hour-last-week rule
5. Pipeline reliability	Fresh, complete, 19 of 19 checks pass; 1 abandoned stage in 30 days